

CEMENT QUALITY CONTROL SYSTEM



25-26J-168

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AGENDA

Problem & motivation

01

Proposed solution overview

02

System Architecture & Integration

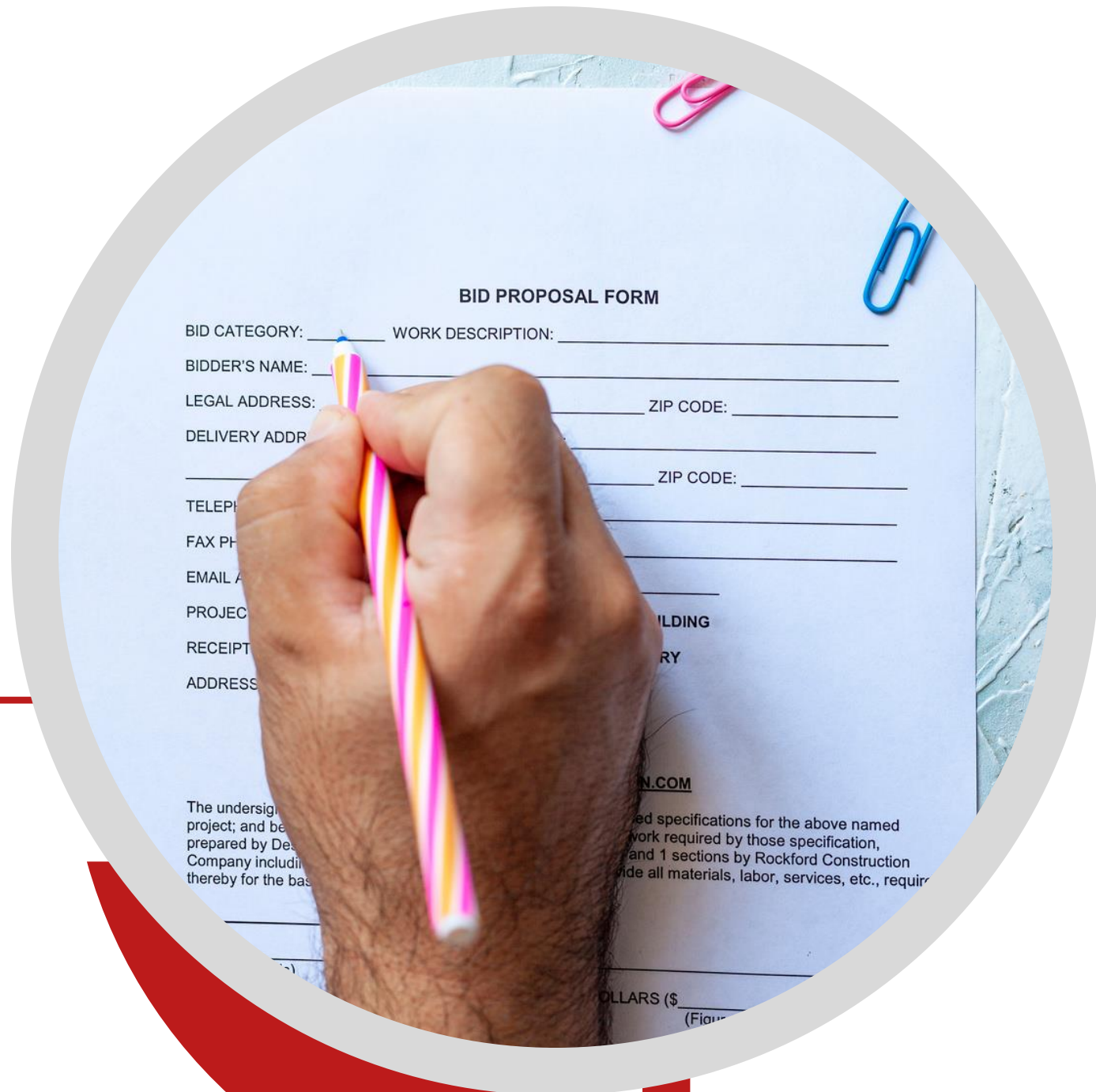
03

Commercialization

04

Demo Plan + roles

05



OUR TEAM



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Koralage K.Y.P
IT22584472

Problem Statement & Stakeholders

- ❖ Manual QC methods → slow (15–60 mins) & expert-dependent
- ❖ Cube testing → Manual measurement, error-prone
- ❖ Traditional testing requires long curing periods
- ❖ Delays cause cost, inconsistency, weak QC



- Stakeholders: Primary: QC engineers, lab technicians
- Secondary: Operators, supervisors
- Decision makers: Quality & plant managers

Research Gap & Objectives

DESIGN EXCELLENCE



Gap

- No integrated, real-time AI for Sri Lankan OPC
- Foreign datasets, ignore local conditions/environment
- Single-task tools only (no 4-in-1 platform)



Main Objective

Build unified AI web platform →
automate 4 QC processes → ~99%
time reduction, higher accuracy, real-
time decisions for local OPC

Solution Overview

❖ Summary: 4-in-1 AI web platform for cement QC automation

Key Features:

- Clinker phase classification (MobileNetV2)
 - Raw meal particle detection & counting (YOLOv9)
 - Cube area + crack detection (U-Net + IoT)
 - Early/long-term strength prediction (XGBoost + LightGBM)
-
- Non-functional: Fast inference (<400ms), secure, user-friendly, reliable
 - Status: ~90%+ complete (models + integration + testing done)



CEMENT CLINKER PHASE CLASSIFICATION



01

Clinker Classification – Abeywickrama A.C.A - IT22640284

02

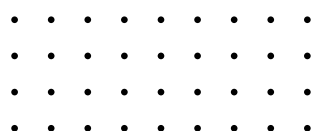
Raw Meal Analysis – Koralage K.Y.P - IT22584472

03

Cube Testing (IoT + U-Net) – Lakshan V.G.S. - IT22562210

04

Strength Prediction – Weerasooriya K.H.P - IT22368034

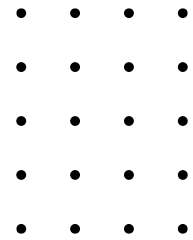


CEMENT QUALITY CONTROL SYSTEM

COMMERCIALIZATION



- ❑ **Target Audience:** INSEE QC engineers and lab staff in cement plants.
- ❑ **Uniqueness:**
 1. AI-based web tool for fast, accurate clinker analysis.
 2. 4-in-1 platform vs single-purpose competitors
- ❑ **Market Size:** Growing demand across the global cement industry for automated QC. (*Our target: Rs. 400-600M in QC automation*)



THANK YOU!

